AFC–10678

Sub. Code
1BCE1C1

B.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Computer Science

PROGRAMMING IN C AND DATA STRUCTURES

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. How do you strings as input and output in 'C'?

2. What are left shift and right shift operation?

3. How to initialize one dimensional array?

4. What is pointer?

5. Define union.

6. What is sequential file?

7. What are the advantages of linked list?

8. How are elements processed in queue?

9. Define degree of node in tree.

10. Define binary tree.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Explain break and continue statement with an example.

Or

(b) Tabulate Arithematic, logical and relational operators supported by 'C'.

12. (a) Write a 'C' program to illustrate passing pointers to a function.

Or

(b) What are the operations done on pointers? Explain.

13. (a) Explain self referential structure.

Or

(b) Explain Enum data type.

14. (a) Explain operation on stack.

Or

(b) Explain operation on queue.

15. (a) How are binary trees represented using array?

Or

(b) How are binary trees represented using linked list?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain storage classes supported by 'C'.

17. Write 'C' program to find out average of given set of numbers using function.

18. Explain operations on a file.

19. Discuss algorithms used to insert a node into the linked list.

20. Discuss applications of binary trees.

AFC–10679

Sub. Code
1BCE2C1

B.Sc. DEGREE EXAMINATION, APRIL 2018.

Second Semester

Computer Science

PROGRAMMING IN C++ AND ALGORITHMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is an object?

2. What is the advantages of inline functions?

3. What is the restriction for static member function?

4. What is the characteristic of court member function?

5. Define constructor?

6. What is the use of destructor?

7. What is the advantage of inheritance?

8. Define virtual functions.

9. What is the principle used in binary research?

10. What is minimum spanning tree?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Compare switch statement with if...else if statement.

Or

(b) Explain rules for coding for...loop.

12. (a) How is memory allocated for static and dynamic objects?

Or

(b) How are objects passed as function argument?

13. (a) Explain about parameterized constructor with an example.

Or

(b) Explain about dynamic initialization of objects.

14. (a) Write a C++ program to overload binary + operator.

Or

(b) Write a C++ program to over load uniary – operator.

15. (a) Explain insertion sort with an example.

Or

(b) Explain merge sort with an example.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain features of object oriented progrmming.

17. Explain friend function with an example.

18. Write a C++ program to generate Fibnoacii series using copy constructor.

19. Explain multilevel and hierarchical inheritance with suitable example.

20. Explain Dijikstra's algorithm.

AFC–10680

Sub. Code
1BCE3C1

B.Sc. DEGREE EXAMINATION, APRIL 2018

Third Semester

Computer Science

DATABASE MANAGEMENT SYSTEMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define DBMS.

2. Define strong and weak entity.

3. What is atomic domain?

4. Define BCNF.

5. What is meant by homogeneous database?

6. What are different types of network?

7. What is role?

8. What is synonym?

9. What is the difference between function and trigger?

10. What is explicit cursor?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain different type of database users.

Or

(b) Explain aggregation in E-R diagram with an example.

12. (a) Explain steps involved in database design process.

Or

(b) Discuss steps involved in modeling temporal data.

13. (a) Explain centralized database system architecture.

Or

(b) Explain inter query parallelism.

14. (a) Write short notes on data integrity.

Or

(b) Explain insert, update and delete operations in tables.

15. (a) Explain stored procedure with an example.

Or

(b) Explain function with an example.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain purpose of database system.

17. Explain decomposition with multivalued dependency.

18. Write short notes heterogeneous databases and distributed transactions.

19. Write short notes on index, sequence and views.

20. Explain about of cursor with an example.

AFC–10681

Sub. Code
1BCE4C1

B.Sc. DEGREE EXAMINATION, APRIL 2018

Fourth Semester

Computer Science

JAVA PROGRAMMING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a Browser?

2. What is JVM?

3. What do you mean by operator precedence?

4. What are the bitwise operators available in java?

5. What is a Constructor?

6. What do you mean by Method Overriding?

7. Define package.

8. What are the advantages of Thread?

9. What is an Applet?

10. Write the syntax of the method which is used to draw an arc.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the features of OOP.

Or

(b) Explain Type casting in java.

12. (a) Explain switch statement with an example.

Or

(b) Write a Java program to print the prime numbers between 150 and 500. Using while loop.

13. (a) How will you implement an interface? Explain with an example.

Or

(b) Explain static members with an example.

14. (a) Explain the Thread priority with example.

Or

(b) Explain the exception handling in java with an example.

15. (a) Explain the Applet tag with an example with all options.

Or

(b) How will you display numerical values in an Applet? Explain with an example.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain command line argument with a program.

17. Write a Java program to find the roots of a quadratic equation for all cases.

18. Write a Java program to find the sum of odd numbers and even numbers separately in an array.

19. How will you create and import a package? Explain with a program.

20. Write a Java program to draw a circle inside a rectangle and display your university name in different background and foreground colors.

AFC–10682

Sub. Code
1BCE5C1

B.Sc. DEGREE EXAMINATION, APRIL 2018

Fifth Semester

Computer Science

OPERATING SYSTEMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is an Operating System?

2. What do you mean by Virtual memory?

3. What is the need for job scheduling?

4. What is Starvation?

5. What are the advantages of Concurrent Programming?

6. What is a dedicated device?

7. Define File.

8. What is a Distributed OS?

9. What are the design goals of UNIX Operating System?

10. What is NOS?

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Explain Fixed partition method with diagram.

Or

(b) Write short notes on Demand Paging.

12. (a) Explain the strategies for handling Deadlock.

Or

(b) Explain any two cases of Deadlock.

13. (a) Explain the Direct Access storage devices.

Or

(b) Explain the components of I/O subsystem.

14. (a) Explain the various File Access methods.

Or

(b) Write short notes on File Organization.

15. (a) Explain the UNIX device management system.

Or

(b) Write short notes on User Interface.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain various types of Operating System.

17. Explain :

(a) SJF

(b) Round Robin Scheduling algorithms with example.

18. Describe the process synchronization software.

19. Explain :

    (a) DOS development

    (b) NOS development.

20. Explain the UNIX File Management system.

AFC–10683

Sub. Code
1BCE5C2

B.Sc. DEGREE EXAMINATION, APRIL 2018

Fifth Semester

Computer Science

COMPUTER ORGANIZATION AND ARCHITECTURE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the uses of Register Transfer Language?

2. List the Logic Micro Operations.

3. What are the Input – Output Instructions?

4. Write the basic principle of Double Precission Addition.

5. What is meant by Reverse Polish Notation?

6. Write the Assembly Language code using three Address Instruction format for $x = (A + B) * (C + D)$.

7. Draw the flow chart for adding and substracting.

8. What is meant by Divide Over Flow?

9. What is meant by Hit ratio and Miss ratio?

10. Draw the Block diagram of Associative Memory.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Explain about the Instruction Cycle.

Or

(b) Explain about the Register Transfer Bus.

12. (a) Describe with Micro Operations of any two memory reference instructions.

Or

(b) Describe about Program Loops.

13. (a) Explain about the Reduced Instruction Set Computer.

Or

(b) Explain about any two Addressing Modes in detail.

14. (a) Draw the Flow Chart for Fixed Point division.

Or

(b) Explain about the Input – Output Processor in detail.

15. (a) Explain about any two auxiliary Memory devices.

Or

(b) List and briefly explain the characteristics of Multiprocessor.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain about the Computer Registers available in the digital Computer.

17. Describe about Subroutines.

18. Explain about Stack Organization in detail.

19. Explain with block diagram DMA data transfer.

20. Discuss in detail about Memory Hierarchy.

———————

AFC–10684

Sub. Code
1BCEE1A

B.Sc. DEGREE EXAMINATION, APRIL 2018

Fifth Semester

Computer Science

Elective — DATA MINING AND DATA WAREHOUSING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is meant by clean and transform data?

2. What is data warehouse?

3. What is capacity planning?

4. What is the need for turning and testing?

5. What is meant by knowledge discovery in database?

6. What are the metrics used in datamining?

7. Define OLAP.

8. What is web search engine?

9. Define association rule.

10. What are itemsets?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain extract and load process.

Or

(b) Explain query management process.

12. (a) Why are tools needed to manage a data warehouse?

Or

(b) Explain steps involved in estimating the load.

13. (a) Discuss data missing issues in brief.

Or

(b) Discuss data mining from a database perspective.

14. (a) What is dimensional modeling? Explain.

Or

(b) Write short notes on fuzzy sets and fuzzy logic.

15. (a) Explain about incremental rules.

Or

(b) Explain measuring the quality of rules.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain process architecture in brief.

17. Discuss about tuning the data load and tuning queries.

18. Explain social implication of data mining.

19. Write short notes on decision trees and genetic algorithm.

20. Explain about advanced association rule techniques.

———————

AFC–10685

Sub. Code
1BCEE2A

B.Sc. DEGREE EXAMINATION, APRIL 2018

Fifth Semester

Computer Science

Elective – WEB DESIGN TECHNOLOGY

(CBCS – 2011 onwards)

Time : Three Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the differences between Markup Language and other High level Programming Languages.

2. Write the Syntex code for connect the HTML form with Back end Server side Programming.

3. What are the Functions of Style Sheets?

4. How will you set to background for a web page?

5. Write the differences between Javascript and VB script.

6. What are the Logical Operators used in Javascript?

7. How will you declare array in Javascript?

8. What is meant by Recussion?

9. Write the Syntex for onclick event.

10. Write the Syntex for DTD.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Write the HTML code for a Single, Simple webpage.

Or

(b) How special characters are inserted in a web page?

12. (a) Write code for create a drop down menu.

Or

(b) Explain the media types.

13. (a) Write the syntex and explain the 'for' structure.

Or

(b) Write the syntex and explain the 'Switch' structure.

14. (a) Explain the programmer defined functions.

Or

(b) Explain the Document object in Java script.

15. (a) Explain the use of XML namespaces with Syntex.

Or

(b) Describe the mouse over the mouse out events.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. How will you create the lists in a web page?

17. Explain the inline and Embedded Style sheets.

18. Explain the increment and decrement operators with example.

19. Explain the window object with a program.

20. Describe the XML schema with a program.

AFC–10686

Sub. Code
1BCE6C1

B.Sc. DEGREE EXAMINATION, APRIL 2018

Sixth Semester

Computer Science

COMPUTER NETWORKS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the components are needed to construct a computer network?

2. Write the evoluationary development of a computer network.

3. What is meant by guided transmission?

4. What are the services of the physical layer?

5. What are the types of errors occurs in data link layer?

6. What is meant by parity checking?

7. Write the need for network layer.

8. What is meant by multiplexing and de-multiplexing?

9. What are mail Access protocols?

10. What is meant by name space?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain about OSI reference model briefly.

Or

(b) What are the applications of computer networks?

12. (a) Explain about fibre optic cable.

Or

(b) Describe about circuit switching in detail.

13. (a) Describe about forward error correction methods.

Or

(b) Describe about Ethernet.

14. (a) Explain the addressing principles in network layer.

Or

(b) Explain about the static versus dynamic routing.

15. (a) Explain about the RSA algorithm.

Or

(b) Describe about world wide web.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe about network standardization in detail.

17. Describe about public switched telephone network.

18. Describe about channel allocation problem in detail.

19. Describe about user Data Gram protocol.

20. Write the technical concepts of Electronic mail.

———————

AFC–10688

Sub. Code
1BCE6C3

B.Sc. DEGREE EXAMINATION, APRIL 2018

Sixth Semester

Computer Science

SOFTWARE ENGINEERING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the categories of Software Products based on size.

2. What are the factors that Influence quality and productivity of a product?

3. Which principle is used in Delphi Cost Estimation?

4. What are the Desirable Properties of Requirements specification document?

5. What are the design notations? Write the names of all.

6. What is meant by Milestone in Software Engineering?

7. Write the principles of System Testing.

8. What is meant by Acceptance Testing?

9. Define : SQA Plan.

10. What is meant by Software Review?

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the skills required for a Software Engineer.

Or

(b) Explain the factors to be consider in setting project goals.

12. (a) Explain the Delphi cost estimation technique.

Or

(b) Explain about the state-oriented notations.

13. (a) Describe about Fundamental Design Concepts.

Or

(b) Explain about Inspections.

14. (a) Explain the Configuration Management.

Or

(b) Describe about the source Code Metrics.

15. (a) Explain the ISO 9000 quality standards.

Or

(b) Explain the Quality Concepts.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the Phased Life – Cycle Model.

17. Describe the Relational notations in detail.

18. Explain about the Structured Flow Chart and Structured English.

19. Explain about the Maintenance tools.

20. Describe about Static Analysis.

AFC–10687

Sub. Code
1BCE6C2

B.Sc. DEGREE EXAMINATION, APRIL 2018

Sixth Semester

Computer Science

COMPUTER GRAPHICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define vector.

2. How to find out distance between a point and a line?

3. What is the difference between random and raster scan display devices?

4. How is polygon represented?

5. What is homogeneous co-ordinate?

6. What is the structure of segment table?

7. Define windowing.

8. What is view port?

9. What is Echoing?

10. What is event?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) What is a line segment? What are its attributes?

Or

(b) Explain about frame buffer.

12. (a) Explain Display-File structure.

Or

(b) Explain An Inside test.

13. (a) Explain the operation of creating and closing of segment.

Or

(b) Explain the operation of renaming and visibility of segment.

14. (a) Derive viewing transformation matrix.

Or

(b) Explain about generalized clipping.

15. (a) Explain about hardware and input device for interaction.

Or

(b) Explain about any one of the interactive techniques.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain about Bresenham's line drawing algorithm.

17. Explain a polygon filling algorithm with an example.

18. Derive matrix representation for rotation about an arbitrary point.

19. Explain Sutherland-Hodgman algorithm.

20. Explain about simulating a locator.

AFC–10689

Sub. Code
1BCEE3A

B.Sc. DEGREE EXAMINATION, APRIL 2018

Sixth Semester

Computer Science

***Elective* — MOBILE COMMUNICATION**

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the applications of mobile communication?

2. What is meant by Multi-path propagation?

3. What is meant by SDMA technology?

4. What are the functions of Mobile services switching center?

5. What are the disadvantages of wireless LAN?

6. Which principle is used in wireless ATM?

7. What is meant by cellular IP?

8. What is meant by snooping TCP?

9. Define : Rover.

10. Define : Ficus.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Describe about Antennas.

Or

(b) Explain about the modulation in a transmitter.

12. (a) Describe about Classical Aloha and Slotted Aloha.

Or

(b) Describe the GPRS architecture reference model.

13. (a) Explain about BRAN.

Or

(b) Explain about Location Management.

14. (a) Explain Dynamic Host Configuration Protocol.

Or

(b) Describe about Indirect TCP in detail.

15. (a) Describe about Wireless Transaction Protocol.

Or

(b) Describe about Wireless Application Protocol.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the frequencies used for the Radio transmission.

17. Describe about DECT system architecture reference model.

18. Describe the Bluetooth technology.

19. Explain about the transaction oriented TCP.

20. Explain about the Hyper Text Markup Language.

———————

AFC–10690

Sub. Code
1BCEE3B

B.Sc. DEGREE EXAMINATION, APRIL 2018.

Sixth Semester

Computer Science

Elective — VISUAL BASIC

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is implicit declarations?

2. Define function.

3. What are the three possible states of a form?

4. Write any four properties of Listbox control.

5. Write a note on Picture Box control.

6. Write the Width and Height properties.

7. What is the use of Treeview and Listview controls?

8. List any two uses of RichTextBox control.

9. What are the three types of Record sets?

10. List the four buttons of Data control.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain if...then...else statement with an example.

Or

(b) Explain Named Arguments with sample coding.

12. (a) How to build dynamic Forms at Runtime.

Or

(b) Describe briefly the File Controls.

13. (a) Explain the methods for drawing lines and shapes with examples.

Or

(b) Write a VB program to generate Curves.

14. (a) Explain the Font Common Dialog Box.

Or

(b) Explain RTF Language with an application.

15. (a) Explain the structure of the BIBLIO Database.

Or

(b) Explain the Data Environment with the DataGrid Control.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain different types of variables available in Visual Basic.

17. Discuss in detail on manipulating menus at Runtime.

18. Write an elaborate note on Graphics Controls.

19. Discuss about Treeview and Listcontrol.

20. Explain Cursor types and Locking mechanisms.

———————

AFC–10725

Sub. Code
1BCESA1/ 1BCEA1

U.G. DEGREE EXAMINATION, APRIL 2018

Computer Science

Allied — DIGITAL ELECTRONICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the differences between analog computer and digital computers?

2. What are the universal logic gates? Draw the truth table and symbols?

3. Write the technical concept of combinational logic circuit.

4. Write the truth table for the code conversion from BCD to excess-3 code.

5. Draw the Basic flip-flop circuit with NAND gates.

6. What is meant by Ring Counter?

7. What are the four levels of integration?

8. What are the configurations of Programmable Logic Device (PLD)?

9. What are the applications of A/D converters?

10. Define : Accuracy and Resolution of A/D and D/A converters.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain about Radix complement and diminished radix complement.

Or

(b) Explain the conversion between canonical forms.

12. (a) Write the Design procedure of a combinational logic circuit.

Or

(b) Design a Half–Substractor.

13. (a) Explain about the Digital clock.

Or

(b) Explain with circuit diagram of a JK master –slave flip-flop.

14. (a) Explain the positive logic NAND gate and positive logic NOR gates with truth table.

Or

(b) Describe about DTL basic Nand gate.

15. (a) Explain about the counter method of A/D conversion.

Or

(b) Explain about the Binary Ladder circuit.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Do the following

    (a) Using 10's complement substract 72532 – 3250

    (b) Using 10's complement substract 3250 – 72532

    (c) Given the two binary numbers X = 1010100; Y = 1000011, perform

    X – Y and Y – X using 2's complement.

17. Explain about the Programmable Logic Array (PLA) with block diagram.

18. Explain about the Serial transfer of Shift Registers.

19. Describe about Bi-polar transister characteristics.

20. Describe the Simultaneous method of A/D conversion.

———————

AFC–10726

Sub. Code
1BCESA2

U.G DEGREE EXAMINATION, APRIL 2018

Computer Science

Allied : MICROPROCESSORS AND INTERFACING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Draw the flag register configuration 8086 Microprocessor.

2. What are the functions of wait state.

3. What are the functions of XCHG Instruction?

4. Write the operation of MOV BL, CL Instruction and draw the op code format of 8086 Microprocessor.

5. Write the basic principle of interfacing memory with 8088 microprocessor.

6. Draw the circuit diagram for interfacing 7 – segment display 8088 microprocessor.

7. What are the functions of 8272 Floppy disk controller?

8. What are the functions of character counter of 8275 CRT Controller?

9. Draw the register set of 80286 Microprocessor.

10. Write the technical parameters of 80386 Microprocessor.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the status lines of 8086 Microprocessor.

Or

(b) Compare the 8086 and 8088 Microprocessor.

12. (a) Explain the addressing modes of 8088 Microprocessor.

Or

(b) Explain the stack related instructions of 8088 Microprocessor.

13. (a) Describe the 8257 (DMAC) Controller.

Or

(b) Describe the 8284 clock generator.

14. (a) Explain the interfacing techniques of Acoustic couplers.

Or

(b) How will you interface printers?

15. (a) Explain the interrupts of 80286 Microprocessor.

Or

(b) Explain the register organization of 80386 Microprocessor.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the execution unit and Bus Interface unit of 8086 Microprocessor.

17. Explain the Arithmetic Instructions of 8088 Microprocessor.

18. Describe the 8251 (USART) controller.

19. Describe the Keyboard Interfacing.

20. Describe the Signal describtions of 80386 Microprocessor.

———————

AFC–10727

Sub. Code
1BCESA3

U.G. DEGREE EXAMINATION, APRIL 2018

Computer Science

Allied — RESOURCE MANAGEMENT TECHNIQUES

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Discuss the role of OR study in industry.

2. Define model.

3. Define feasible solution.

4. Give the matrix form of an LPP.

5. State the optimality condition in the dual simplex method.

6. State the fundamental theorem of duality.

7. Write the mathematical formulation of an Assignement problem.

8. Distinguish between transportation model and assignment model.

9. Briefly explain the transportation problem.

10. What is an unbalanced transportation problem?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) What is operation research? Describe briefly its applications.

Or

(b) Describe the features of OR.

12. (a) Solve graphically the LPP

Maximize $z = 7x_1 + 3x_2$

Subject to $x_1 + 2x_2 \geq 3$

$$x_1 + x_2 \leq 4$$
$$0 \leq x_1 \leq 5/2$$
$$0 \leq x_2 \leq 3/2$$

and $x_1, x_2 \geq 0$.

Or

(b) Use the simplex method to solve the LPP

Maximize $z = 10x_1 + x_2 + 2x_3$

Subject to $x_1 + x_2 - 2x_3 \leq 10$

$$4x_1 + x_2 + x_3 \leq 20$$

and $x_1, x_2, x_3 \geq 0$.

13. (a) Apply the dual simplex method to solve

Minimize $z = 4x_1 + 2x_2$

Subject to the constraints

$$3x_1 + x_2 \geq 27$$
$$x_1 + x_2 \geq 21$$
$$x_1 + 2x_2 \geq 30$$

and $x_1, x_2 \geq 0$.

Or

(b) Write the step-by-step procedure to solve an IPP using branch and bound method.

14. (a) Write the Hungarian algorithm to solve an Assignment problem.

Or

(b) Solve the following Assignment problem.

		Jobs			
		J_1	J_2	J_3	J_4
Workers	W_1	10	15	24	30
	W_2	16	20	28	10
	W_3	12	18	30	16
	W_4	9	24	32	18

15. (a) What is degeneracy in the transportation model? How it can be overcome?

Or

(b) Obtain an initial basic feasible solution of a TP using the row minima method.

	A	B	C	
I	50	30	220	1
II	90	45	170	3
III	250	200	50	4
	4	2	2	

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the scope of operations research.

17. Solve using Big-M method.

Maximize $z = 3x_1 - x_2$

Subject to the constraints

$2x_1 + x_2 \geq 2$; $x_1 + 3x_2 \leq 3$; $x_2 \leq 4$ and $x_1, x_2 \geq 0$.

18. Solve the IPP

Maximize $z = x_1 + 2x_2$

Subject to the constraints

$x_1 + x_2 \leq 7$; $2x_1 \leq 11$; $2x_2 \leq 7$ $x_1, x_2 \geq 0$ and are integers.

19. Solve the following traveling salesman problem.

	A	B	C	D	E	F
A	∞	5	12	6	4	8
B	6	∞	10	5	4	3
C	8	7	∞	6	3	11
D	5	4	11	∞	5	8
E	5	2	7	8	∞	4
F	6	3	11	5	4	∞

20. Solve the following transportation problem.

	D_1	D_2	D_3	D_4	D_5	Supply
O_1	3	2	3	4	1	100
O_2	4	1	2	4	2	125
O_3	1	0	5	3	2	75
Demand	100	60	40	75	25	300

AFC–10728

Sub. Code
1BCESA4

U.G. DEGREE EXAMINATION, APRIL 2018

Computer Science

Allied : COMPUTER ORIENTED NUMERICAL METHODS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is a Bisection method?

2. Define Back Substitution.

3. What is method of Least Square?

4. Define Eigen vectors of a matrix.

5. What is interpolation?

6. Write Lagrange's Interpolation formula.

7. Write the first and second derivatives of central difference formula.

8. Write Gaussian Quadrative formula.

9. Define Euler's method.

10. Write Milne's predictor–corrector formula.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Find the root of the equation $x^3 - 2X - 5 = 0$ using Regula Falsi Method.

Or

(b) Explain Horner's method.

12. (a) Explain the linear law for curve fitting.

Or

(b) Find the Eigen values of the matrix $\begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

13. (a) Form a table of backward differences of the function $f(x) = x^3 - 3x^2 - 5x - 7$ for $x = -1, 0, 1, 2, 3, 4, 5$

Or

(b) Construct a table if divided difference for the following data :

x	0	2	3	5	6
y	1	19	53	241	415

14. (a) Derive Newton's forward difference formula to compute the derivatives.

Or

(b) Evaluate $\int_0^1 \frac{1}{1+x} dx$ correct to three decimal places by trapezoidal rule with $h = 0.5$.

15. (a) Explain Taylor's series method.

Or

(b) Solve the equation $\frac{dy}{dx} = 1 - y$ with the initial condition $x = 0, y = 0$. Using Euler's method and tabulate the solutions at $x = .1, .2, .3, .4$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Using Crout's method, solve the system of equations

$2x - 6y + 82 = 24$

$5x + 4y - 32 = 2$

$3x + y + 2z = 16$

17. Find a straight line fit of the for m $y = a + bx$, by the method of group averages for the following data

x	50	70	100	120
y	12	15	21	25

18. Apply a central difference formula to obtain $f(32)$ that

$f(25) = 0.2707,\ f(30) = 0.3027\ f(35) = 0.3386,$

$f(40) = 0.3794$.

19. Evaluate $\int_0^1 \frac{1}{1+x}\,dx$ correct to three decimal places by Romberg's method using $h = 0.25$.

20. Apply Runge Kutta method of fourth order to find an approximate value of y when $x = 0.2$ given $y' = 3x + \frac{1}{2}y,\ y(0) = 1$.

AFC–10786

Sub. Code
1BCAE2A

B.C.A DEGREE EXAMINATION, APRIL 2018

Fifth Semester

Computer Application

Elective : COMPUTER GRAPHICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. List out the domains or area where computer graphics are applied.

2. What are the advantages of Bresenham's line drawing algorithm?

3. What are basic transformations?

4. What is the advantage of matrix representation?

5. Define Aspect Ratio.

6. Define view port.

7. How do move an object in 3D?

8. What is matrix reflection transformation?

9. What is meant by styles of command language?

10. What is the need for user interface?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain characteristics and functions of anyone graphics input device.

Or

(b) Explain Ellipse drawing algorithm.

12. (a) Derive 2D composite transformation matrix.

Or

(b) Derive 2D matrix representation for translation.

13. (a) Explain about viewing transformation.

Or

(b) Explain line segment clipping algorithm.

14. (a) Derive 3D rotation transformation matrix.

Or

(b) Derive 3D translation transformation matrix.

15. (a) Write short notes on command language.

Or

(b) Explain about information display in user interface design.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain DDA line drawing algorithm.

17. Derive 2D scaling and rotation transformation matrix..

18. Explain Sutherland Hodgman clipping algorithm.

19. Derive 3D composite transformation matrix and explain its advantages.

20. Explain the components of user model in detail.